

conditional statements

```
In [1]: if True:  
        print('Hello sennapati sir teaching was excellent')
```

Hello sennapati sir teaching was excellent

```
In [2]: if False:  
        print('your assumption was wrong')
```

```
In [3]: if True:  
        print('Datascience')  
  
        print('bye')
```

Datascience
bye

```
In [4]: if False:  
        print('Datascience')  
  
        print('bye')
```

```
In [5]: if True:  
        print('Data Science')  
  
        else:  
            print('bye for now')
```

Data Science

```
In [6]: if True:  
        print('Data Science')  
  
        else:  
            print('bye for now')
```

Data Science

```
In [7]: if False:  
        print('data science')  
        else:  
            print('bye')
```

bye

A python code to check even or odd number

```
In [8]: x = 4  
        r = x%2  
        if r == 0:  
            print('even number')
```

even number

```
In [9]: x = 3
r = x%2
if r == 0:
    print('even number')
```

```
In [10]: x = 6
r = x%2
if r == 0:
    print('even number')
if r == 1:
    print('odd number')
```

even number

```
In [11]: x = 7
r = x%2
if r == 0:
    print('even number')
if r == 1:
    print('odd number')
```

odd number

```
In [12]: x = 6
r = x%2
if r == 0:
    print('even number')
else:
    print('odd number')
```

even number

```
In [13]: x = 5
r = x%2
if r == 0:
    print('even number')
else:
    print('odd number')
```

odd number

```
In [14]: x = 6
r = x % 2

if r == 0:
    print('Even number')
print('odd number')
```

Even number

odd number

```
In [15]: x= 4
r = x % 2
if r == 0:
    print('even number')
else:
    print('odd number')
```

even number

```
In [16]: x = 5
r = x % 2

if r == 0:    print('Even number')
else:        print('odd number')
```

odd number

```
In [17]: x = 10
r = x % 2
if r == 0:
    print('even number')
if r == 1:
    print('odd number')
```

even number

```
In [18]: x = 9
r = x % 2
if r == 0:
    print('even number')
if r != 0:
    print('odd number')
```

odd number

```
In [19]: x = 10
r = x % 2
if r == 0:
    print('even number')
if r != 0:
    print('odd number')
```

even number

nested if condition

```
In [20]: x = 3
r = x % 2
if r == 0:
    print('even number')
    if x > 5:
        print('greatest number')
else:
    print('odd number')
```

odd number

```
In [21]: x = 6
r = x % 2
if r == 0:
    print('even number')
    if x > 5:
        print('greatest number')
else:
    print('odd number')
```

even number
greatest number

```
In [22]: x = 4
r = x % 2
if r == 0:
    print('even number')
    if x > 5:
        print('greatest number')
    else:
        print('lowest number')
else:
    print('odd number')
```

even number
lowest number

```
In [23]: x = 4
if x == 1:
    print('one')
if x == 2:
    print('two')
if x == 3:
    print('three')
if x == 4:
    print('four')
```

four

```
In [24]: x = 2
if x == 1:
    print('one')
elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')
```

Two

```
In [25]: x = 10
if x == 1:
    print('one')
elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')
```

```
In [26]: x = 10
if x == 1:
    print('one')
elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')
```

```
else:  
    print('number not found')
```

number not found

```
In [27]: num = int(input('enter a number'))  
if num > 0:  
    print("positive")  
elif num < 0:  
    print("negative")  
else:  
    print("zero")
```

positive

LOOPS

```
In [28]: # WHILE LOOP  
i = 1  
while i<=5:  
    print('data science')  
    i = i + 1 # increment
```

data science
data science
data science
data science
data science

```
In [31]: i = 5  
while i >= 1:  
    print('data science')  
    i = i - 1
```

data science
data science
data science
data science
data science

```
In [32]: i = 1  
while i <= 5:  
    print('data science : ', i)  
    i = i + 1
```

data science : 1
data science : 2
data science : 3
data science : 4
data science : 5

```
In [33]: i = 5  
while i >= 1:  
    print('data science : ', i)  
    i = i - 1
```

```
data science : 5
data science : 4
data science : 3
data science : 2
data science : 1
```

```
In [34]: i = 1
        while i <= 5:
            print('data science')
            j = 1
            while j<=4:
                print('technology')
                j = j+1
            i = i + 1
            print()
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
In [36]: i = 1
        while i <= 5:
            print('data science',end=" ")
            j = 1
            while j<=4:
                print('technology',end=" ")
                j = j+1
            i = i + 1
            print()
```

```
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
```

```
In [37]: i = 1
        while i <=2:
            j =0
            while j<=2:
                print(i*j,end=" ")
                j +=1
            print()
            i +=1
```

```
0 1 2
0 2 4
```

```
In [40]: i = 1
        while i <= 4:
            j = 0
            while j<=3:
                print(i*j,end=" ")
                j +=1
            print()
            i = i+1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

foor loop

```
In [41]: name = 'nit'
        for i in name:
            print(i)
```

```
n
i
t
```

```
In [44]: name1 = [1,3,8.9,'nit']
        for i in name1:
            print(i)
```

```
1
3
8.9
nit
```

```
In [45]: for i in [2,4.6,'nit']:
        print(i)
```

```
2
4.6
nit
```

```
In [46]: range(5)
```

Out[46]: range(0, 5)

```
In [47]: for i in range(5):  
         print(i)
```

0
1
2
3
4

```
In [48]: for i in range(1,5):  
         print(i)
```

1
2
3
4

```
In [49]: for i in range(1,10,3):  
         print(i)
```

1
4
7

```
In [50]: for i in range(1,21):  
         print(i)
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
In [52]: # print the value which is divisible by 5  
         for i in range(1,51):  
             if i%5==0:  
                 print(i)
```


5
10
15
20
25
30
35
40
45
50

```
In [53]: # print value which is not divisible by 5  
# print the value which is divisible by 5  
for i in range(1,51):  
    if i%5!=0:  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

```
In [54]: # write a code to ask chocolate from vendor machine  
x = int(input("how many chocolates do you want?"))
```

```
i = 1
while i<=x:
    print('here your chocolate')
    i +=1
```

here your chocolate
here your chocolate
here your chocolate
here your chocolate
here your chocolate

```
In [60]: avail_choco = 5
x = int(input("how many chocolates do you want"))
i = 1
while i<=x:
    if i > avail_choco:
        break
    print("chocolate")
    i=i+1
print("thank you")
```

chocolate
chocolate
chocolate
chocolate
thank you

```
In [61]: avail_choco = 5
x = int(input("how many chocolates do you want"))
i = 1
while i<=x:
    if i > avail_choco:
        break
    print("chocolate")
    i=i+1
print("thank you")
```

chocolate
chocolate
chocolate
chocolate
chocolate
thank you

```
In [62]: avail_choco = 5
x = int(input("how many chocolates do you want"))
i = 1
while i<=x:
    if i > avail_choco:
        print('out of stock')
        break
    print("chocolate")
    i=i+1
print("thank you")
```

chocolate
chocolate
chocolate
chocolate
chocolate
out of stock
thank you

```
In [63]: for i in range(1,11):  
         print(i)
```

1
2
3
4
5
6
7
8
9
10

```
In [64]: for i in range(1,11):  
         if i==6:  
             break  
         print(i)
```

1
2
3
4
5

```
In [65]: for i in range(1,11):  
         if i ==3:  
             continue  
         print(i)
```

1
2
4
5
6
7
8
9
10

```
In [66]: for i in range(1,11):  
         if i == 6:  
             continue  
         print('hello :',i)
```

```
hello : 1  
hello : 2  
hello : 3  
hello : 4  
hello : 5  
hello : 7  
hello : 8  
hello : 9  
hello : 10
```

```
In [67]: for i in range(1,11):  
        pass
```

print 1 to 50 numbers but not divisible by 3 and 5

```
In [69]: for i in range(1,51):  
        if i%3==0:  
            print(i)
```

```
3  
6  
9  
12  
15  
18  
21  
24  
27  
30  
33  
36  
39  
42  
45  
48
```

```
In [70]: for i in range(1,51):  
        if i%3==0:  
            continue  
        print(i)
```

1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50

```
In [71]: for i in range(1,51):  
         if i%3==0 & i%5==0:  
             continue  
         print(i)
```

1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50

```
In [73]: for i in range(1,51):  
         if i%3==0 or i%5==0:  
             continue  
         print(i)  
         print('end')
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end

```
In [76]: # print only odd numbers from 1 to 50  
for i in range(1,51):  
    if(i%2==0):  
        continue  
    else:  
        print(i)  
print('end')
```

```

1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
end

```

PATTERNS IN PYTHON

```

In [77]: print('# # # #')
         print('# # # #')
         print('# # # #')
         print('# # # #')

```

```

# # # #
# # # #
# # # #
# # # #

```

```

In [79]: for i in range(1,5):
         i +=1
         print('# # # #')

```

```

# # # #
# # # #
# # # #
# # # #

```

```

In [81]: for i in range(1,5):
         if i<=5:
             print('# # # #')

```

```

# # # #
# # # #
# # # #
# # # #

```

```

In [82]: for j in range(4):
         print('# # # #')

```



```
# # # #
# # # #
# # # #
# # # #
```

```
In [84]: for j in range(4):
          print('#', end=" ")
```

```
# # # #
```

```
In [85]: for j in range(4):
          print('#', end = " ")
          for j in range(4):
              print('#', end = " ")
```

```
# # # # # # # #
```

```
In [86]: for j in range(4):
          print('#', end=" ")
          print()
          for j in range(4):
              print('#', end=" ")
```

```
# # # #
# # # #
```

```
In [87]: for j in range(4):
          print('#', end=" ")
          print()
          for j in range(4):
              print('#', end=" ")
              print()
          for j in range(4):
              print('#', end=" ")
              print()
          for j in range(4):
              print('#', end=" ")
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [88]: for i in range(4):
          for j in range(4):
              print('#',end=" ")
          print()
```

```
# # # #
# # # #
# # # #
# # # #
```

```
# #
```

```
In [89]: for i in range(4):
          for j in range(i+1):
```

```
print('#',end = " ")
print()
```

```
#
# #
# # #
# # # #
```

In [90]: `for i in range(5):`
`print('# '*i)`

```
#
# #
# # #
# # # #
```

In [94]: `for i in range(1,5):`
`for j in range(4):`
`if i>j:`
`print("#",end=" ")`
`print()`

```
#
# #
# # #
# # # #
```

In [95]: `for i in range(5):`
`for j in range(4):`
`if i>j:`
`print("#",end=" ")`
`print()`

```
#
# #
# # #
# # # #
```

```
# #
```

In [99]: `for i in range(4):`
`for j in range(4-i):`
`print('#',end = " ")`
`print()`

```
# # # #
# # #
# #
#
```

In [121... `for i in range(4): # mirror pattern for above`
`for j in range(i):`
`print(' ',end=" ")`
`for j in range(4-i):`
`print("#",end=" ")`
`print()`

```
# # # #  
# # #  
# #  
#
```

```
In [122... for i in range(5):  
            for j in range(4):  
                if i>j:  
                    print("#",end=" ")  
            print()
```

```
#  
# #  
# # #  
# # # #
```

```
In [ ]:
```